AgentWorkShop: A Configurable Framework for Multi-Scenario Industrial Agent Integration

***Abstract*—Industrial agents need a configurable path from task execution and production context to heterogeneous equipment, with operations scoped and outcomes traceable. We present AgentWorkShop, a modular framework that combines collaborative task services and registered harness interfaces; scene blueprints and line–product–recipe–run context; semantic data acquisition and write control; and analysis and model services. A shared dispatcher and route handlers connect bounded node-exploration and recipe workflows, while authorization, command receipts, process observations, and recipe versions remain distinct records. We evaluate historical simulator and protocol archives, engine/controller/recipe traces, deployment regressions, and a current v0.8.0, two-scene application-level integration run plus a separate registered-harness trial. Across cast-film and annealing, five protocol-driver families passed read probes; bounded writes were verified and 1,393 DAQ samples were collected into lineage-linked datasets. Both process-model candidates failed the configured eligibility gates, so no recommendation or dispatch followed. In the separate trial, the registered adapter and dispatcher carried one manually approved `recipe_trial` per scene to a verified write/readback, followed by one run-tagged DAQ snapshot. These simulator checks demonstrate the tested paths, not model success, process convergence, cross-harness portability, physical-plant transfer, or comparative control benefit.**

***Index Terms*—Industrial agents, industrial communication, industrial informatics, production systems, semantic data integration.**

I. Introduction

LARGE language-model agents are being explored for plant-facing tasks. Xia et al. present an end-to-end industrial-control framework using structured prompts and event-driven production context [1]. Other work studies operator-reviewed control-sequence generation [2], modular factory skills [3], and production scheduling [4]. Together, these studies establish a broad industrial-agent research direction, with different task and integration boundaries. Across plants, agent roles, signals, protocols, objectives, and production context vary. This paper focuses on the software integration problem of configuring those differences around reusable task services, registered harnesses, shared operation checks, semantic process I/O, and route-specific records; it does not claim the first agent-to-control integration.

Consider injection molding. An agent team may use recent measurements and process knowledge to diagnose insufficient packing and propose a holding- pressure change. To carry that proposal into an auditable production task, a system must resolve the active line, product, recipe, and run; map the quantity to the correct device interface; check that the requested operation is available for that context; and retain the command evidence alongside subsequent quality observations. Another process may use different signals, protocols, or objectives while reusing the same collaboration and runtime services. The design problem is therefore to make scene-specific variation configurable while keeping each operation tied to its production context.

This paper presents AgentWorkShop, a configurable industrial-agent framework that connects collaborative tasks and registered harnesses to production context, manufacturing execution system (MES) data, analysis and modeling services, data-acquisition (DAQ) and write-control (DCW) nodes, and protocol-specific drivers. Its access surfaces include a command-line interface (CLI), web, software development kit (SDK), representational state transfer (REST), Model Context Protocol (MCP), and Agent-to-Agent (A2A) interfaces. Scene blueprints define signal maps, objectives, and guards; channel profiles, tool policies, and agent bindings scope operations. Agents and callers invoke named operations through shared runtime checks; the framework does not choose routes autonomously and retains proposals, command confirmation, and process observations as distinct records. Figure 1 presents the system architecture, and Fig. 2 shows the cross-module task flow. Section IV details the node-exploration and recipe lifecycles.

The paper makes three contributions:

1) It formulates five deployment requirements and six design rules for connecting agent collaboration, production context, process interfaces, and route-specific evidence. These are explicit design choices for the target workflows, not a taxonomy claimed to be empirically derived.
2) It specifies AgentWorkShop's cross-module execution flow and software modules, including task and harness integration, production-context configuration, analysis and modeling services, semantic device access, and route-specific evidence handling.
3) It reports complementary evidence from historical simulator and protocol archives, live-engine tasks, a deterministic controller, deployment regressions, and a recipe-path case. A current-version application-level two-scene integration run verifies protocol reads, bounded writes, readback, and data lineage, while its model gates reject both candidate datasets. A separate two-scene trial follows the registered-harness, approval, write, readback, and run-tagged acquisition path using a deterministic RPC fixture. Each evidence unit is tied to its interface and software version; the results do not establish successful optimization, process convergence, multi-harness portability, or comparative control benefit.

Section II reviews related industrial-agent and integration work. Section III presents the design rationale, architecture, and cross-module execution flow; Section IV explains module responsibilities and their integrated operation. Section V de-

TABLE I
SELECTED PLANT-FACING AGENT SYSTEMS AND THEIR INTEGRATION SETTINGS. EXECUTION AND REPORTED EVIDENCE ARE DESCRIBED WITHIN EACH SOURCE'S SCOPE; THE TABLE IS NOT A PERFORMANCE RANKING.

System	Task or execution unit	Industrial integration interface	Operation or reported evidence
Xia et al. [1]	End-to-end production planning and control by an industrial-task agent system	Structured prompts and an event-driven model supplying real-time, multilevel context	Agents generate plans and control operations; proof-of-concept implementation and task-specific dataset method
Köhle et al. [3]	One LLM agent per factory module; comparison of orchestrator, peer-to-peer, and monolithic architectures	Per-module MCP servers expose OPC UA methods; MQTT coordination and real-time factory-state injection	State-based tool filtering precedes method calls; nine simulated production challenges, including silent-fault detection
Biskupovic et al. [5]	Industrial agents select and reconfigure services in service-oriented ICPSs	Predefined component library; selection uses performance metrics and computational load	Online monitoring triggers reselection after poor performance; reported adaptability and performance results
AgentWorkShop (this work)	A collaboration task maps to a registered harness and an industrial operation	Configured line, product, recipe, and run context; semantic I/O resolves through protocol-specific drivers	Shared dispatch checks scope, grants, and bindings; route handlers retain command and process evidence as distinct records

scribes the experiments and results, and Sections VI and VII discuss the evidence limits and conclude the paper.

II. RELATED WORK

This section connects AgentWorkShop to two complementary bodies of work: plant-facing industrial agents, and the production and automation abstractions that define context, tool access, and operational constraints. Table I compares selected agent systems; the research gap is limited to these comparators rather than attributed to the wider literature.

A. Industrial Agents and Plant Integration

Industrial-agent systems span end-to-end automation control, supervisory services, programmable logic controller (PLC) interfaces, manufacturing coordination, and operator-facing machine access. Xia et al. present an LLM-agent framework for end-to-end industrial automation control, using structured prompts and event-driven context for production planning and operation [1]. Biskupovic et al. study agent-based supervision for service-oriented industrial systems [5]; Agents4PLC and InstructMPC address PLC code generation and human–LLM context-aware control, respectively [6], [7]. Manufacturing studies consider dynamic resource-capability exploration [8], multi-agent shop-floor coordination [9], and digital-twin-supported fault handling [10]. Natural-language machine access [11] and real-time LLM control of a robot-operated serial line [4] extend this work to operator interfaces and robotic-line control. Together, these studies address different operational contexts, interfaces, and allocations of responsibility between agents, operators, and controllers. These comparators overlap with distinct parts of the integration problem: end-to-end LLM control, module-level factory skills, and service-oriented supervision. Table I compares their execution units, industrial interfaces, and source-reported execution or follow-up. The comparison is purposive and descriptive; it does not score system capabilities or compare performance.

The three systems therefore organize execution around different units: automation-system tasks for Xia et al., factory modules for Köhle et al., and services within an ICPS for Biskupovic et al. [1], [3], [5]. AgentWorkShop instead defines a configurable task-to-operation integration contract: a collaboration task resolves to production scene/run context and a registered harness; returned calls pass through shared dispatch checks before route-specific handlers access semantic process nodes through protocol drivers. Command receipts, process observations, and recipe versions remain separate records. This contribution concerns integration and execution contracts, not a new control or service-selection algorithm. Current evidence covers the framework components and a two-scene protocol/data path, but does not combine the registered-harness-to-process path in one trial. This scope boundary neither implies that the cited systems lack similar mechanisms nor establishes performance superiority. Authority and command confirmation are operational services, consistent with safety-aware decision support [12]; they do not replace process protection or establish operator effectiveness.

B. Production Context, Agent Interfaces, and Automation

Industrial integration also depends on abstractions that are not specific to agent architectures. MCP standardizes tool discovery and calls with request-scoped authorization [13], whereas ISA-88 and ISA-95 describe recipes and production operations [14], [15]. Run-to-run methods connect process observations to successive production runs [16]. At the agent level, AgentSpec and CaMeL address tool behavior and capability-based information flow [17], [18]. Together, these works provide complementary abstractions for production context, tool interaction, and information flow. They do not, by themselves, specify the shared integration architecture presented here.

Industrial security and record requirements further constrain that architecture: FDA guidance discusses trustworthy records and audit trails [19], IEC 62443-3-3 specifies system-security requirements [20], and DNP3 includes select-before-operate [21]. Control-assurance methods address state evolution through shielding, control-barrier functions, or Simplex architectures [22]–[24]. These standards and methods inform the boundaries within which an industrial-agent framework

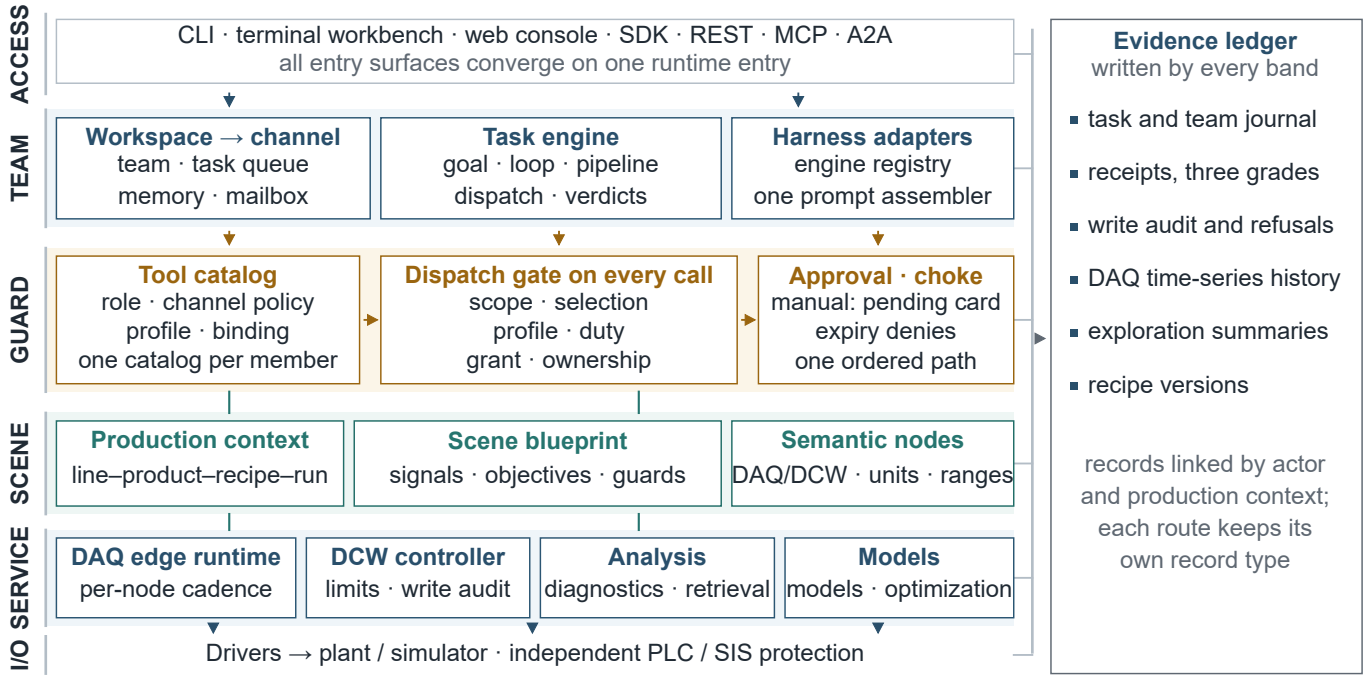


Fig. 1. AgentWorkShop software architecture. Access, collaboration, governance, production context, services, and field I/O form six connected bands. Registered harness and driver interfaces expose extension points; the evidence ledger retains route-specific records across the bands. The field interface ends at the registered drivers; PLC and safety-instrumented protection remain external.

must operate; they are not contributions claimed by AgentWorkShop.

Table I provides a purposive, descriptive comparison, not a performance ranking. AgentWorkShop's contribution is implementation-level integration of registered harnesses, scoped production context, shared dispatch, semantic process I/O, and route-specific evidence. The evaluation does not validate per-agent profile adaptation, a new control strategy, or performance superiority.

III. System Design

AgentWorkShop is designed for industrial tasks whose agents, production contexts, interfaces, and process signals vary between deployments. A useful framework must let these differences be configured without changing the shared task runtime, while ensuring that a proposed operation is checked against the active production context before it reaches a device. It must also retain enough route-specific evidence to distinguish a returned command receipt from a measured process response.

A. Design Rationale and Requirements

The five requirements—scene variability, harness variability, operation authority, observability, and extensibility—organize six design rules (D1–D6). They are deployment constraints selected from the target workflows, not a taxonomy inferred from a systematic review or an empirical coding study: encode scene settings as versioned data; extend the runtime through registered harness, tool, and driver interfaces; separate proposal from execution; treat the tool catalog as discovery rather than permission; route actuator writes through the common DCW admission point; and retain evidence by operation. D1 maps to blueprints and context resolution; D2 to the harness, tool, and driver registries; D3–D5 to task dispatch, binding/grant checks, and DCW write admission; and D6 to route handlers and the evidence ledger. The evaluation maps these rules to complementary evidence: Q1 and Q4 exercise historical direct-node scene mappings; Q2 exercises two tasks with one registered OMP harness; Q5 checks dispatcher and write-confirmation regressions on v0.7.57; Q6 and the October 23 boundary record exercise recipe and approval paths; and Q7 checks current v0.8.0 protocol acquisition, bounded writes, readback, and data lineage across two simulator scenes. A separate Q8 trial traverses the registered harness through approval, write/readback, and one run-tagged acquisition in both scenes, but does not test an LLM decision or process convergence. Q7 and Q8 are separate evidence units, not a single combined deployment trial. The results state available version and interface provenance and identify gaps, including the unavailable R1 runtime worktree and R6 commit. AgentWorkShop does not select an agent's control strategy or replace independent controller and safety functions.

B. Layered Architecture

Fig. 1 maps the reference implementation onto six bands and a shared evidence ledger. The access band exposes command-line, web, SDK, REST, MCP, and A2A entry surfaces. The collaboration band contains workspaces, channels, task queues, shared memory, mailboxes, task templates, and harness adapters. The governance band projects an operation catalog and checks each invocation. The context band binds a scene blueprint to the active line–product–recipe–run tuple and its semantic DAQ/DCW nodes. The service band provides acquisition, write control, analysis, retrieval, and optional

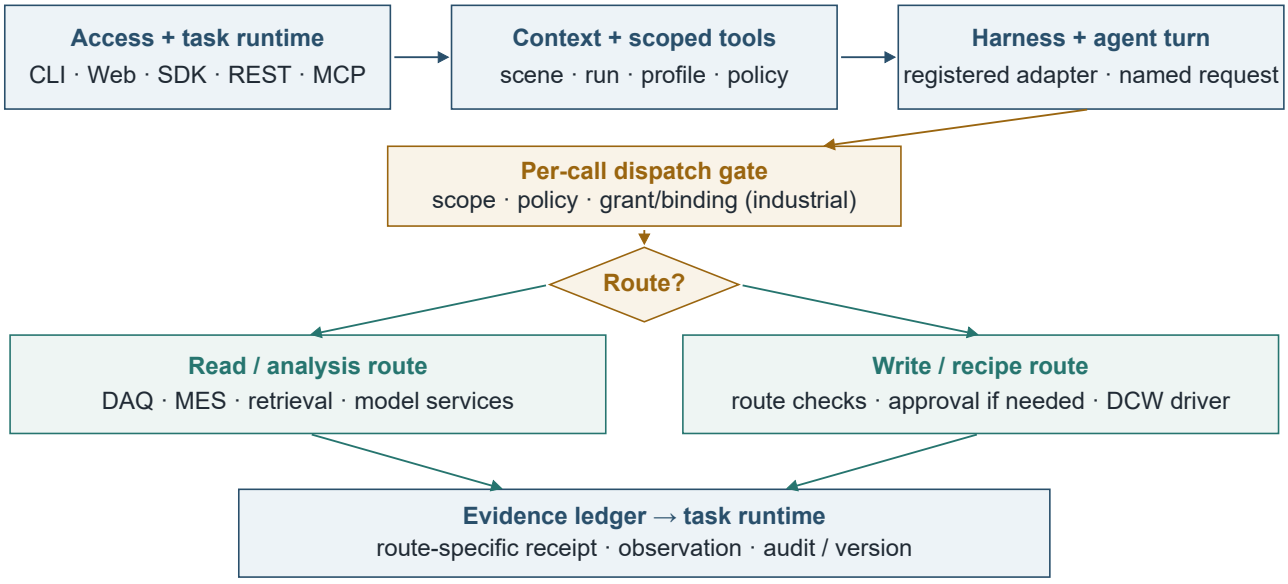


Fig. 2. Cross-module task execution in AgentWorkShop. Entry and task services assemble the channel-scoped tool catalog before the agent turn; each returned call is rechecked for scope and policy, with grant and binding checks for industrial operations. Dispatch-gate refusals stop before handler execution, whereas route-specific handlers may read state before refusing a write. Receipts and later observations remain distinct.

model services. The field-I/O band resolves semantic nodes to built-in or plugin drivers for a plant or simulator.

Each band has a distinct responsibility. Entry surfaces converge on the same task runtime; a harness adapter supplies the agent-specific interaction; semantic nodes hide protocol details from the task; and the driver registry provides the device-facing extension point. The evidence ledger records task, command, observation, and version events against their actor and production context. It does not collapse them into a single success flag.

C. Cross-Module Execution Flow

Fig. 2 summarizes a task turn from entry to returned evidence. The runtime resolves the production context and assembles the prompt and scoped tool catalog before invoking a registered harness. For each returned tool call, the shared dispatcher rechecks invocation scope and channel policy; industrial operations also receive current grant and binding checks before handler selection. The selected handler applies its own route-specific guards, approval, revalidation, and device-I/O steps. Command receipts and any later process observations remain distinct records.

Fig. 3 gives pseudocode for these cross-module contracts; it is a reference flow rather than one literal orchestrator function. Route-specific timing, approval, and evidence rules are detailed in Section IV.

Require: task q, channel c, bound agent a
Ensure: task result and route-linked evidence
$x \leftarrow \textsc{ResolveContext}(c, q)$
$T \leftarrow \textsc{FilterToolCatalog}(a, c)$
$m \leftarrow \textsc{RetrieveMemory}(q)$
$p \leftarrow \textsc{AssemblePrompt}(q, x, m)$
$h \leftarrow \textsc{InvokeRegisteredHarness}(a, p, T)$
for all tool request u returned by h **do**
  $g \leftarrow \textsc{CheckScopeAndChannelPolicy}(a, c, u)$
  if not g **then**
    $\textsc{RecordDispatchRefusal}$(u,x); **continue**
  end if
  if u is an industrial operation **then**
    $v \leftarrow \textsc{CheckGrantAndBinding}(a, u)$
    if not v **then**
      $\textsc{RecordDispatchRefusal}$(u,x); **continue**
    end if
  end if
  $k \leftarrow \textsc{SelectHandler}(u)$
  $r \leftarrow \textsc{ExecuteRouteHandler}(k, u, x)$
  $\textsc{RecordAvailableEvidence}$(u,r,x)
  $\textsc{ReturnToolResult}$(h,r)
end for
return task state and retained evidence

Fig. 3. Reference pseudocode for cross-module execution in AgentWorkShop.

IV. Module Functions and Integration

The architecture in Fig. 1 is implemented as cooperating modules rather than a single monolithic agent. Shared interfaces connect task execution to configuration, industrial services, protocol drivers, and route-specific evidence. This section describes those responsibilities and the checks that preserve their boundaries.

The extension contracts expose the data passed at each boundary. A harness call is normalized as `HostToolCall(toolName, arguments)` and returns `HostToolResult(text, isError)`. A DCW driver

accepts an engineering value, tolerance, range, and driver configuration; its result separates write status, raw value, readback, and acknowledgment grade (readback-verified, transport-ack, or unverified). Each DAQ sample stores its node, timestamp, value, and state, with optional line, product, recipe, and run IDs. A write anchor records line/node, prior and requested values, source, actor, time, and optional run/task/approval IDs; recipe definitions retain separate versions. These fields keep tool errors, device receipts, observations, and production lineage inspectable as separate records.

A. Task Runtime and Harness Adapters

An access surface creates or joins a workspace channel, where the task engine manages members, task queues, shared memory, and execution templates. The harness registry instantiates an adapter behind a common agent interface; the prompt builder combines the task with the active scene brief, industrial context, channel profile, agent binding, and retrieved memory. Tool calls return to the shared dispatcher regardless of which registered harness produced them. The registry is an extension point; the experiments do not establish portability across harnesses.

B. Production Context and Configuration

Scene blueprints describe devices, signal maps, objectives, and process guards. A task resolves these definitions against its active line–product–recipe–run context. The channel profile selects the available capability family, the tool policy narrows the operation catalog, and the agent binding scopes the nodes, controls, and recipe operations available to that member. These filters can restrict one another but cannot broaden the underlying device range. The dispatcher re-reads policy at execution, so a cached catalog does not retain authority after a grant or policy change.

C. Acquisition, Analysis, and Modeling Services

Semantic DAQ nodes expose typed values, engineering units, ranges, and sampling settings while hiding protocol-specific reads. A per-node DAQ runtime schedules acquisition and stores timestamped observations; the gateway aggregates node pipelines. MES-facing operations provide the configured production-data path. Analysis and retrieval services consume these records, while optional model, training, optimization, and digital-twin services are invoked through their registered operation handlers. These services complement the agent workflow; AgentWorkShop does not prescribe a single optimization strategy.

D. Semantic I/O and Actuation

The driver registry resolves a semantic node to a built-in or plugin driver. DAQ handlers use the node's acquisition interface; DCW handlers route a write or readback through the controller and the configured protocol driver. This separation lets recipes and agents refer to semantic controls without embedding Modbus, OPC UA, MQTT, HTTP, or MES transport details in their operation arguments. The supported driver mappings are implementation interfaces; only the mappings and workflows enumerated in Section V are experimentally exercised.

E. Shared Dispatch and Write Admission

Every named operation passes the dispatcher. Its ordered checks validate invocation scope, channel tool selection, profile, duty separation, the agent's binding and runtime grant, and ownership of referenced resources. A refusal returns the excluding condition and is recorded. A catalog entry therefore describes availability but does not authorize execution. Handlers for operations that require manual approval suspend for a decision, then revalidate authority and active limits before dispatch.

All actuator writes converge on the DCW admission point. For an online node write, the admissible interval is

$$\mathcal{I}_n^{\text{online}}(t) = \mathcal{H}_n \cap \mathcal{B}_n \cap \mathcal{P}_n(t) \cap \mathcal{R}_n(t), \tag{1}$$

where $\mathcal{H}_n$ is the hard engineering range and $\mathcal{B}_n$, $\mathcal{P}_n$, and $\mathcal{R}_n$ are the baseline, active-product, and active-recipe limits. Missing optional limits impose no restriction, whereas an empty intersection rejects the request and identifies the excluding layer. Recipe dispatch applies the hard, baseline, and product limits to the replacement recipe rather than intersecting with the previous recipe window. The actuator path also enforces step, interval, and hold-window checks. These admission rules constrain commands; process objectives and coupled variable guards are assessed from later observations. The framework does not replace independent PLC or safety-instrumented protection.

F. Integrated Operation and Route-Specific Evidence

A recipe trial illustrates how the modules cooperate. The task runtime delivers a named `recipe_trial` call from the agent harness; the dispatcher checks the agent's recipe binding and execution scope; the recipe handler checks the active run, candidate parameters, and route timing; and a manual binding requests approval before the DCW controller sends the batch through its driver. The handler returns a per-parameter receipt. The caller then obtains DAQ observations and can request a separate update, apply, or rollback operation. A trial receipt confirms command handling; it does not by itself establish process attainment or persist a new recipe version.

The routes retain different records because they answer different questions. A driver acknowledgment reports command evidence; DAQ reports a process observation; an exploration summary records a bounded probe; and a recipe version records a saved definition. Their relation through actor and production context allows an execution to be inspected without treating these outcomes as interchangeable.

G. Route-Specific Lifecycles and Evidence

a) Timing and approval: Timing guards are route-specific. Online node writes enforce an effective minimum

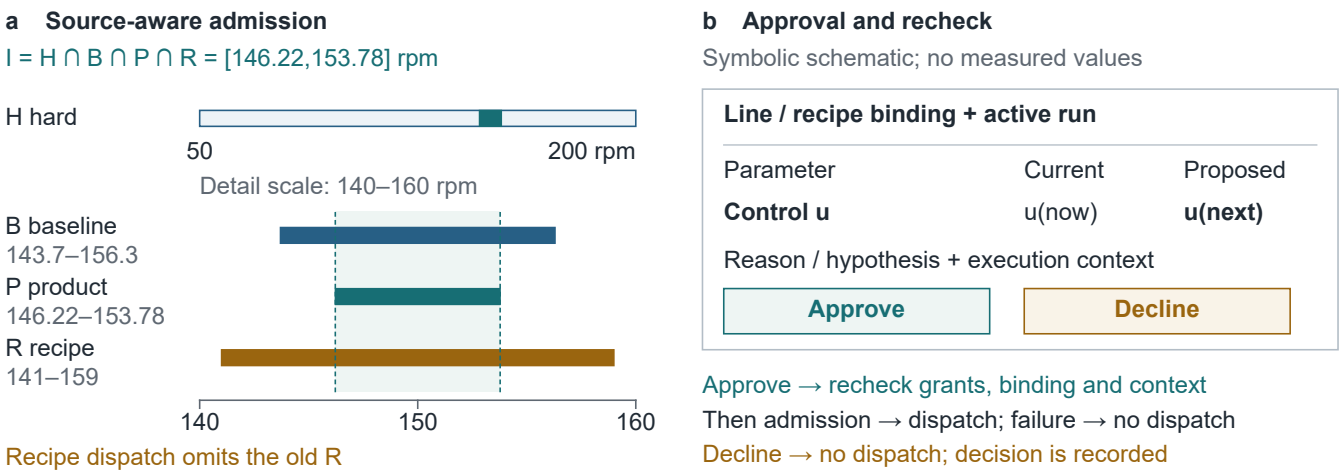


Fig. 4. Admission and approval boundaries. (a) Archived Q1 RPM example: online node interventions use the four-layer value interval of Eq. (1), whereas recipe dispatch omits the prior recipe window. (b) Symbolic human approval binds a proposed action and authority fields; execution checks run again after approval, which does not reserve equipment state.

spacing of 60 s. Recipe dispatch uses a per-recipe interval anchored to approval, while trials also use a configurable per-line cadence (default 300 s). The recipe interval can refuse a proposal before an approval request is created. Checks occur at proposal or execution according to the operation, and experiment-specific pacing overrides are reported with the corresponding results.

Manual bindings create an approval request containing the actor, node or recipe, candidate values, applicable limits, expiry, and parameter bases. The default wait expires after 180 s and refuses execution. After approval, the executor rechecks authorization and active limits to account for context changes during the wait. For `recipe_apply`, a changed definition requires a new proposal; an unchanged approval dispatches a copy of the approved parameter snapshot. A line-level recipe mode defaults to manual; changing it to automatic requires administrator confirmation. The approval record authorizes a workflow action and does not replace process protection.

b) Node probe: The bounded node probe is exposed as `optimization_explore` to the Auto Modeling Lab (AML) optimization profile (`aml_optimization`). The profile validates a bound control and target DAQ scope, captures a starting control and target baseline, invokes the internal DCW write path, then waits 10–300 s (default 45 s) and queries a trailing 45-s target window before saving an exploration summary. Since each window is relative to its query time, approval can separate baseline sampling from actuation; a 10-s wait can leave about 35 s of pre-write samples in the later window. The command receipt and target summary are therefore returned separately. A successful changed-value write opens a node-optimization record; refusal or expired approval creates a rejected summary without a post-write sample.

c) Recipe operations: `recipe_trial` checks the bound recipe and active run, then dispatches candidate overrides as a batch with per-parameter verification. The caller subsequently queries DAQ and decides whether to save a new definition with `recipe_update`, restore one with `recipe_rollback`, or dispatch a version with `recipe_apply`. These steps are separate: a trial does not save or apply a version, and endpoint verification is not a process-response measurement. Public direct-node control is closed; the bounded probe and recipe operations are the configured operation surfaces. The historical four-scene results use an earlier direct-node path and do not test current profile or binding adaptation across scenes.

d) Command verification: Command confirmation and process response answer different questions. A driver ACK is graded as `readback-verified`, `transport-ack`, or `unverified`. A matching endpoint read supports the first grade; a transport or MES application receipt supports the second. Recipe batch verification assigns `verified`, `unverified`, or `failed`; if no matching readback is available, it attempts up to three independent reads, 2 s apart. A matching retry upgrades the final ACK, while exhausted retries remain unverified. The initial ACK remains in write history and the final ACK and verdict are retained with the batch. Per-parameter receipts preserve successful outcomes when another parameter fails. A readback-verified setpoint does not establish target attainment.

e) Observation, verdict, and compensation: The caller performs recipe-path observation after batch dispatch. The October recipe case therefore compares a recipe definition, endpoint readback, and DAQ sample as separate representations. Saving a definition, applying it to equipment, and observing its process response are separately evidenced operations. The evaluation does not combine these route-specific records into one universal end-to-end record.

An open node record provides a monitoring backstop. After 120 s it polls every 30 s; three cumulative samples outside the recipe monitoring window trigger a system-attributed rollback request toward the preceding value. Missing samples do not count as breaches. A caller's `keep`, `rollback`, or `uncertain` verdict records its interpretation; compensation is a new command subject to its applicable admission policy. The write audit deduplicates the same value on the same node within 10 s, while batch receipts, exploration summaries, DAQ history, and recipe versions retain their distinct lifecycles.

Linking these records by actor and production context allows the command, endpoint evidence, and process observation to be inspected without treating them as interchangeable.

V. Experimental Evaluation

The evaluation asks whether the framework connects configured process scenes through the current software stack, and what the archived process, protocol, and route-level evidence establishes beyond that integration check. It combines a current-version two-scene application-level simulator run and a separate registered-harness trial, alongside historical process trajectories, protocol exercises, live-engine tasks, a deterministic controller, deployment regressions, and a recipe case. These evidence units are not matched trials or a paired strategy comparison. Q1–Q6 retain their historical archive identifiers; Q7 denotes the current-version application-level run, and Q8 the separate registered-adapter trial. These labels identify evidence groups rather than posed research questions. All process experiments use repository simulators. Q7 and Q8 identify separate current-version evidence groups. The reproducibility notes map each claim to its source and interface version.

A. Evidence Units and Experimental Procedure

Q1–Q4 use the seed-42 integrated archive (R1) and its raw scenario records; the admission ablation uses a separate companion run (R2). R1 can be audited, but its historical runtime worktree is not archived, so the recorded source commit does not guarantee exact replay. These historical missions use the direct-node interface and scripted proposal policies, apart from the two Q2 live-engine tasks. Q3 additionally uses seeds 43–44; Q5 uses nine deployment-suite executions, and Q6 uses the October 18 recipe archive. The cast-film reference preset uses an author-defined objective J combining thickness conformance, defects, energy, and throughput.

The same 65 profile/composition checks passed on October 20, 24, and 25 with temporary storage and mock I/O; these repeated software checks are not 195 independent assertions or process trials. Q4 retains one seed-42 trajectory per scene, with sequential writes and post-wait observations.

Q5 covers 20 deployment phases in nine historical v0.7.57 runs; its 782 assertions are software checks, not current-version or plant results. Six writes per protocol line yield an upper median and observed maximum, not a tail percentile.

B. Current-Version Two-Scene Simulator Integration (Q7)

An isolated local v0.8.0 application ran the cast-film and annealing simulator presets. The five configured protocol-driver families—OPC UA, MQTT, Modbus TCP, Modbus RTU over TCP, and HTTP/REST—passed read probes, and scene-specific DCW writes returned matching readbacks. Six 120-s runs per scene at 1-s sampling yielded lineage-verified datasets (Table II); neither the simulator truth/optimum endpoint nor an external LLM service was used.

Eligibility used paired row values: MV at the end of the history window and CV at the start of the forecast horizon. After first differences, the service selected the signed Pearson correlation between ΔMV and lagged ΔCV with greatest absolute magnitude over row lags 1–3. This configured gate tests only that three-row window; a failed pair does not establish the absence of a delayed physical response outside it.

This differenced-series gate statistic is distinct from the dataset builder's descriptive raw-level correlation from the first training run over ± 20 lags (see `REPRODUCIBILITY.md`). The configured gates required $|r_\Delta| \geq 0.2$, one-step test normalized root-mean-square error (NRMSE) ≤ 0.10, and at least a 20% reduction in four-step rollout NRMSE relative to persistence, defined as $1 - \mathrm{NRMSE}_{\mathrm{rollout}}/\mathrm{NRMSE}_{\mathrm{persistence}}$. Both candidates were rejected, with no profile, recommendation, or dispatch. Each test split used two of the six 120-s runs (220 film rows and 218 annealing rows); the overlapping 1-s windows are serially dependent, not independent trials. The reported signal pairs followed an exploratory pilot and inspection of preset mappings; both pilot models also failed. Q7 therefore demonstrates the current protocol/data path and gate behavior, not confirmatory model performance. The pilot and final-run data are retained separately in the evidence supplement.

C. Registered-Harness Process Path (Q8)

An isolated v0.8.0 application ran one deterministic RPC fixture turn through the registered `omp` adapter, dispatcher, manual recipe approval, recipe handler, and DCW in each scene. The fixture emitted one predeclared `recipe_trial` (no LLM or agent decision): screw speed 120 to 125 rpm via OPC UA, and zone-3 setpoint 730 to 735°C via Modbus TCP. A synthetic administrator approved each; both writes returned 1/1 verified readback. After 20 s, one explicit probe/ingest yielded one run-tagged DAQ value per scene (46.37 μm; 674.799988°C). This confirms the configured harness-to-simulator path, not a trajectory, settled response, quality improvement, or portability; no recipe was saved.

D. Historical Four-Process Simulator Missions (Q4)

The concurrent campaign provides one seed-42 trajectory per process. Figure 5 pairs archived manipulated controls u with the corresponding response y rather than showing response alone. All four terminal attainment checks passed; two lay near a target boundary.

Injection molding used two changes: hold pressure $45 \rightarrow 58.2$ bar and hold time $6 \rightarrow 10.64$ s. Part weight rose from 31.25 to 32.17 g for a 32.5 ± 0.35 g window, only 0.02 g above its lower limit. Biaxially oriented polyethylene terephthalate (BOPET) changed casting speed $32 \rightarrow 33.8$ m/min, fast-roll speed $118 \rightarrow 120.3$ m/min, and rail width $3000 \rightarrow 3035$ mm. Thickness fell from 28.10 to 25.66 μm for a 25.0 ± 0.7 μm target, 0.04 μm inside the upper limit. Its three changes each have a recorded keep verdict and journal identifier.

Wastewater treatment required 15 writes across several nodes. Dissolved oxygen rose from 0.21 to 3.50 mg/L, within 3.4 ± 0.6 mg/L. Some admissible blower proposals reduced

TABLE II
CURRENT V0.8.0 TWO-SCENE SIMULATOR INTEGRATION (ONE DATASET PER SCENE; SIX SEQUENTIAL 120-S RUNS PER DATASET). COUNTS ARE DAQ POINTS, DATASET ROWS, AND HELD-OUT TEST ROWS. EACH TEST SPLIT USES TWO RUNS PER SCENE; OVERLAPPING 1-S ROWS ARE SERIALLY DEPENDENT, NOT INDEPENDENT REPLICATES.

Scene	Manipulated and controlled signals	DAQ points / rows / test rows	Eligibility and held-out result
Cast film	Screw speed (OPC UA) → thickness (MQTT)	697 / 658 / 220	Blocked; $\lvert r_\Delta \rvert = 0.095$ (lag 2 rows), one-step NRMSE $= 0.259$, four-step reduction $= 14.1\%$
Annealing	Zone-3 temperature setpoint (Modbus TCP) → strip temperature (Modbus TCP)	696 / 655 / 218	Blocked; $\lvert r_\Delta \rvert = 0.053$ (lag 3 rows), one-step NRMSE $= 0.363$, four-step reduction $= 4.8\%$

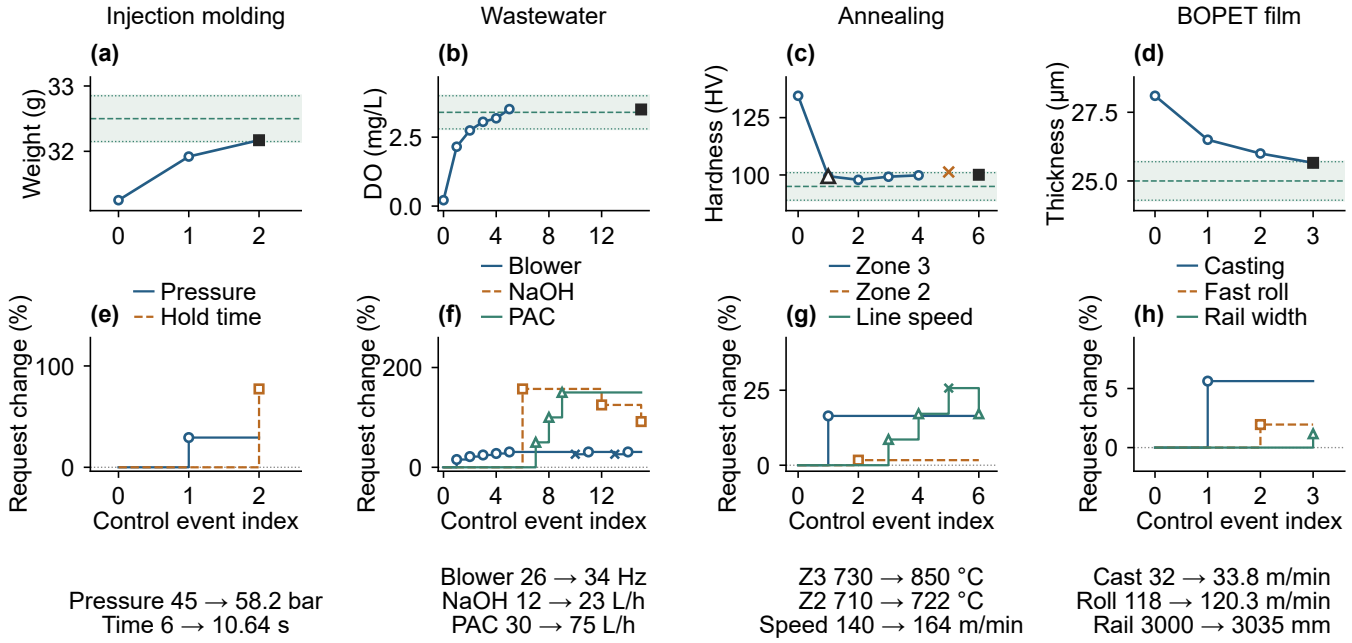


Fig. 5. Historical scripted missions (seed 42, one trajectory per process). (a–d) Weight, dissolved oxygen (DO), hardness, and film thickness with their target windows. (e–h) Requested controls as percentage change from native baselines: injection pressure 45 bar/time 6 s; wastewater blower 26 Hz/NaOH 12 L/h/PAC 30 L/h; annealing zone 3 730°C/zone 2 710°C/speed 140 m/min; BOPET casting 32 m/min/fast roll 118 m/min/rail 3000 mm. Lines connect event indices, not physical time; unchanged requested values carry forward. Crosses mark reverted candidates and squares distinct terminal observations. The open triangle in panel (c) marks the first annealing observation, which lacks an optimization-record ID. DO excludes pH/TP events.

compliance margin and were reverted. Terminal guards remained within their declared limits, although pH moved toward its lower bound (6.59 to 6.38; limit 6.0). Dissolved oxygen, total phosphorus (TP), and pH are different quantities and are not concatenated into one output trajectory.

Annealing combined furnace-temperature changes with line-speed increases from 140 to 164 m/min. The recorded Vickers hardness (HV) trajectory moved from 134.52 to 99.84 HV; the separate terminal measurement was 100.08 HV, within 95 ± 6 HV. A 176 m/min attempt produced 101.3 HV and was reverted. The first temperature step has no optimization record identifier in the archive, which bounds record-level traceability. These historical missions document process behavior under a shared scripted task scaffold and retain the distinction between a candidate command, a reverted candidate, and a terminal observation. They do not evaluate current profile or per-agent binding variation across scenes.

E. Heterogeneous Protocol Integration and Route Checks (Q1)

The suite acquired 55 samples over five protocol families and issued policy-checked writes over four; the RTU-over-TCP line was acquisition-only. Archived upper-median write latencies ranged from 19.3–47.8 ms, with observed maxima 20.1–248.9 ms. The four writable lines accepted legal writes and rejected 24/24 out-of-window probes. A configuration-only second preset added 9/9 rejected probes on three further writable lines. These results exercise the same resolver against different signal maps.

In a mock-driver API fixture, three deterministic repetitions of six attack classes per arm isolated admission from endpoint confirmation. The full and no-readback arms each rejected all 18 probes. The no-interlock and ungated arms each rejected the 12 hard-range probes but allowed all six soft-window probes to execute. Thus removing readback did not change admission, while removing the soft-window gate changed it; confirmation cannot replace a pre-write gate. A frozen-measurement drill further separated command and process evidence: the

TABLE III
FINITE-GRID SEARCH PROTOCOL FOR THE DETERMINISTIC CAST-FILM CONTROLLER IN Q3. THE REFERENCE SCORE IS RESTRICTED TO THIS GRID AND IS NOT A GLOBAL CONTINUOUS OPTIMUM.

Search element	Candidate set or rule	Role in the search
Objective	Eq. (2); weights 55/25/8/7 for thickness, quality, energy proxy, and throughput proxy	Rank eligible steady states
Eligibility	Melt temperature 195–225 °C; melt pressure $\leq$ 22 MPa	Discard ineligible states
Equal zone setpoint	195–225 °C in 2.5 °C steps	Search variable
Screw speed	50–200 rpm in 5-rpm steps	Search variable
Line speed	20–120 m/min in 2.5-m/min steps	Search variable
Die gap	Fixed at 1 mm	Held constant
Selection	Evaluate modeled steady state and retain the highest eligible score; $J^* = 89.894$	Finite-grid reference

TABLE IV
RECORDED Q6 INJECTION-RECIPE ACTIONS AND POST-WAIT OBSERVATIONS ($n = 1$ SIMULATOR TRAJECTORY). HOLD TIME REMAINED 6 S; EACH TRIAL USED HARNESS APPROVAL, A 65-S WAIT, AND A 1/1 PARAMETER-VERIFICATION RECEIPT.

Stage	Pressure (bar)	Weight (g)	Flash (%)	Sink (%)	Recorded action and outcome
Baseline	45	31.28	0	1.977	No command; initial observation
Trial 1	45→50	31.49	0.014	1.956	Approved; 1/1 verified; continue
Trial 2	50→55	31.74	0.031	1.760	Approved; 1/1 verified; continue
Trial 3	55→60	32.05	0	1.534	Weight below 32.15-g minimum and sink above 1.50%; continue
Trial 4	60→65	32.30	0.019	1.277	Stop recorded; displayed values are within ranges
Save	65	—	—	—	`recipe_update` saved version 2; no `recipe_apply` is archived

setpoint and readback agreed, yet the monitoring quantity remained outside its window and triggered backstop rollback after 130.2 s. A manual-binding drill suspended execution until harness-issued approval (461 ms mediation span).

F. Live-Engine Tasks and Deterministic Controller (Q2–Q3)

The archived team workflow linked parent task, worker, command, verdict, and journal. Its single-node cast-film task completed after changing the heater from 200 to 204.704 °C and observing 204.70 °C within target. Two live-engine tasks used the same `omp` harness and GLM-5.3-Flash provider on a scratch line. Their logs trace a bounded write, descent to 51.7 μm for a $52.0 \pm 0.8\,\mu$m target, and recipe save; the 52.02 μm window mean differs from the run-summary terminal value. These traces demonstrate execution integration, not strategy comparison or cross-harness portability.

A deterministic controller used the dimensionless cast-film objective

$$J = 55J_{\text{thick}} + 25J_{\text{quality}} + 8J_{\text{energy}} + 7J_{\text{throughput}}, \tag{2}$$

whose components score thickness conformance, quality, and energy/throughput proxies. The latter use screw and line speed, not metered energy or output. This deterministic controller is not the optional model service. Table III gives its finite grid; formulas and replay details are in `REPRODUCIBILITY.md`.

Across seeds 42–44, the controller made two policy-checked writes per seed without rejection and raised J from about 67.2 to $J/J^* = 0.965$–0.972 ($J^* = 89.894$ on the admissible finite grid). This demonstrates one conventional strategy through the shared execution services, not comparative control benefit.

G. Deployment Composition and Confirmation Semantics (Q5)

The archived v0.7.57 deployment suite composed 34 lines, 111–113 channels, 67 online DAQ nodes, 77 write nodes, 52 recipes, and 385 bindings on one simulator host. Setpoint/readback/mirror values agreed within 0.0 bar, but S4 remained `converged=false`; 65-bar observation-to-verification took 96.105 s. S5 applied 40-to-40.5°C, refused a subsequent write, and restored 40°C after 66.046 s. Seven injected faults exercised receipt grading, partial batch failure, and manual confirmation. These software checks characterize only the tested faults and do not estimate field rates.

H. Operational Recipe Workflow Case Study (Q6)

R6 is one October 18 injection-simulator trajectory with four approved 5-bar `recipe_trials` from 45 bar, each followed by a 65-s wait. It saved recipe version 2 but did not apply it (Table IV); the source commit, seed, and per-sensor timestamps are unavailable, and the intervention was scripted. R6 bypassed trial cadence and the separate operation interval. Melt temperature was missing; stopping used weight, flash, and sink, with freshness based on the newest available timestamp. Thus the recorded stop is not an independently audited guard pass. A separate October 23 public-route check refused eight direct-node writes; its default-cadence recipe run missed target and failed restoration. Neither it nor the distinct 65-to-70 bar check replicates R6.

VI. DISCUSSION

Q7 tests the current protocol/data path, but both model candidates fail the configured gates. Q8 completes one fixture-issued, manually approved, readback-verified recipe trial per scene and one run-tagged DAQ snapshot. This bounded integration check does not establish an autonomous decision, process response, convergence, or quality improvement. Historical Q4 used a scripted direct-node interface; profile checks used mock I/O. Current profile/binding adaptation, cross-harness portability, and physical transfer remain untested.

Q6 saved but did not apply a recipe; its cadence-bypassed stop does not prove simultaneous freshness, and a separate

default-paced check missed target and failed restoration. Approval records do not measure operator effectiveness [25]–[27]; restricted tool access does not establish deployment security [20], [28], [29] or replace independently engineered PLC/SIS protection [30].

Further evaluation needs a versioned build, matched scenes, a predeclared signal-pair/excitation plan, default-paced recipe baselines, and physical plant trials.

VII. Conclusion

AgentWorkShop connects collaborative harnesses, production context, model services, semantic I/O, and node/recipe workflows while retaining distinct authorization, command-confirmation, and observation records. Current simulator evidence verifies selected integration paths; Q7's model gates reject both candidates, and Q8 completes one bounded harness-driven trial per scene. Optimization success, process convergence, cross-harness portability, recipe recovery, plant transfer, and comparative control benefit remain unestablished.

References

[1] Y. Xia, N. Jazdi, J. Zhang, C. Shah, and M. Weyrich, "Control industrial automation system with large language model agents," in *Proc. IEEE 30th Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205539.

[2] K.-C. Huang, H.-T. Lin, C.-H. Lin, and M.-T. Huang, "Large language model integration in industrial cyber-physical systems: A multi-modal architecture for automated control generation," in *Proc. Int. Conf. on Control, Automation and Diagnosis (ICCAD)*, 2026, pp. 1–6, doi: 10.1109/ICCAD69956.2026.11643124.

[3] K. Köhle, D. Anicic, T. A. Runkler, and R. Graf, "LLM-driven multi-agent control for skill-based smart manufacturing," in *Proc. IEEE 31st Int. Conf. on Emerging Technologies and Factory Automation (ETFA)*, 2026, pp. 1–8, doi: 10.1109/ETFA68377.2026.11725257.

[4] M. Waseem, K. Bhatta, C. Li, and Q. Chang, "Pretrained LLMs as real-time controllers for robot operated serial production line," *Mach. Learn.: Eng.*, vol. 2, no. 1, 2026, art. no. 015006; doi: 10.1088/3049-4761/ae4ee8.

[5] A. Biskupovic, A. Villalonga, F. Castaño, R. E. Haber, and F. Núñez, "Agent-based supervision for service-oriented industrial cyber–physical systems," *IEEE Trans. Ind. Inform.*, vol. 21, no. 3, pp. 2719–2728, 2025, doi: 10.1109/TII.2024.3514178.

[6] Z. Liu *et al.*, "Agents4PLC: Automating closed-loop PLC code generation and verification in industrial control systems using LLM-based agents," *IEEE Trans. Softw. Eng.*, vol. 52, no. 5, pp. 1672–1687, 2026, doi: 10.1109/TSE.2026.3667895.

[7] R. Wu, J. Ai, and T. Li, "InstructMPC: A human-LLM-in-the-loop framework for context-aware control," in *Proc. 2025 IEEE 64th Conference on Decision and Control (CDC)*, 2025, pp. 172–179, doi: 10.1109/CDC57313.2025.11312156.

[8] J. Lim and I. Kovalenko, "A large language model-enabled control architecture for dynamic resource capability exploration in multi-agent manufacturing systems," in *Proc. IEEE 21st Int. Conf. on Automation Science and Engineering (CASE)*, 2025, pp. 2088–2095, doi: 10.1109/CASE58245.2025.11163802.

[9] Z. Zhao *et al.*, "A large language model-based multi-agent manufacturing system for intelligent shopfloors," *Adv. Eng. Inform.*, vol. 69, 2026, art. no. 103888; doi: 10.1016/j.aei.2025.103888.

[10] M. S. Gill, J. Vyas, A. Markaj, F. Gehlhoff, and M. Mercangöz, "Leveraging LLM agents and digital twins for fault handling in process plants," in *Proc. IEEE ETFA*, 2025, pp. 1–8, doi: 10.1109/ETFA65518.2025.11205597.

[11] B. Hofmann, N. Piechulek, S. Kreitlein, J. Franke, and P. Bruendl, "Beyond touch-based human-machine interface: Control your machines in natural language by utilizing large language models and OPC UA," *arXiv preprint arXiv:2510.11300*, 2025. [Online]. Available: https://arxiv.org/abs/2510.11300v2

[12] X. Jiang *et al.*, "Agentic AI for safety-aware process monitoring and fault diagnosis: A review," *Processes*, vol. 14, no. 13, 2026, art. no. 2112; doi: 10.3390/pr14132112.

[13] Model Context Protocol contributors, "Model Context Protocol: Tools," Specification, version 2026-07-28, 2026, accessed: Oct. 20, 2026. [Online]. Available: https://modelcontextprotocol.io/specification/2026-07-28/server/tools

[14] ISA, "ISA-88.00.01-2010: Batch control—part 1: Models and terminology," International Standard, 2010.

[15] ——, "ANSI/ISA-95.00.01-2025 (IEC 62264-1 mod): Enterprise-control system integration—part 1: Models and terminology," International Standard, 2025.

[16] J. Moyne, E. del Castillo, and A. M. Hurwitz, Eds., *Run-to-Run Control in Semiconductor Manufacturing*, 1st ed. Boca Raton, FL: CRC Press, 2000.

[17] H. Wang, C. M. Poskitt, and J. Sun, "AgentSpec: Customizable runtime enforcement for safe and reliable LLM agents," in *Proc. IEEE/ACM 48th Int. Conf. on Software Engineering (ICSE)*, 2026, pp. 2938–2950, doi: 10.1145/3744916.3764546.

[18] E. Debenedetti *et al.*, "Defeating prompt injections by design," 2025, arXiv preprint arXiv:2503.18813. [Online]. Available: https://arxiv.org/abs/2503.18813

[19] U.S. Food and Drug Administration, "Part 11, electronic records; electronic signatures—scope and application," Guidance for Industry, 2003.

[20] IEC, "IEC 62443-3-3:2013: Industrial communication networks—network and system security—part 3-3: System security requirements and security levels," Standard, 2013.

[21] IEEE, "IEEE std 1815-2012: Standard for electric power systems communications—distributed network protocol (DNP3)," Standard, 2012.

[22] M. Alshiekh, R. Bloem, R. Ehlers, B. Könighofer, S. Niekum, and U. Topcu, "Safe reinforcement learning via shielding," in *Proc. AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 2669–2678, doi: 10.1609/aaai.v32i1.11797.

[23] A. D. Ames, X. Xu, J. W. Grizzle, and P. Tabuada, "Control barrier function based quadratic programs for safety critical systems," *IEEE Trans. Autom. Control*, vol. 62, no. 8, pp. 3861–3876, 2017, doi: 10.1109/TAC.2016.2638961.

[24] L. Sha, "Using simplicity to control complexity," *IEEE Softw.*, vol. 18, no. 4, pp. 20–28, 2001, doi: 10.1109/MS.2001.936213.

[25] R. Parasuraman and D. H. Manzey, "Complacency and bias in human use of automation: An attentional integration," *Hum. Factors*, vol. 52, no. 3, pp. 381–410, 2010, doi: 10.1177/0018720810376055.

[26] R. Parasuraman, T. B. Sheridan, and C. D. Wickens, "A model for types and levels of human interaction with automation," *IEEE Trans. Syst., Man, Cybern., Part A*, vol. 30, no. 3, pp. 286–297, 2000, doi: 10.1109/3468.844354.

[27] L. Bainbridge, "Ironies of automation," *Automatica*, vol. 19, no. 6, pp. 775–779, 1983, doi: 10.1016/0005-1098(83)90046-8.

[28] K. Greshake, S. Abdelnabi, S. Mishra, C. Endres, T. Holz, and M. Fritz, "Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection," in *Proc. 16th ACM Workshop on Artificial Intelligence and Security*, 2023, pp. 79–90, doi: 10.1145/3605764.3623985.

[29] Y. Liu, Y. Jia, R. Geng, J. Jia, and N. Z. Gong, "Formalizing and benchmarking prompt injection attacks and defenses," in *Proc. 33rd USENIX Security Symposium (USENIX Security 24)*, 2024, pp. 1831–1847. [Online]. Available: https://www.usenix.org/conference/usenixsecurity24/presentation/liu-yupei

[30] IEC, "IEC 61511-1:2016: Functional safety—safety instrumented systems for the process industry sector—part 1: Framework, definitions, system, hardware and application programming requirements," International Standard, 2016.